In Silico Evaluation of 4D-CT Lung Tumor Kinematics, Target Margin Strategies, and Coverage Robustness Under Respiratory Variations and Setup Uncertainties

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ABSTRACT

Background: Accurate respiratory motion management is vital in lung stereotactic body radiation therapy (SBRT). Conventional static 3D-CT expansions frequently either underdose moving lesions or excessively irradiate healthy lung parenchyma. This study establishes an open-source in silico simulation platform evaluating 4D-CT respiratory kinematics, internal target volume (ITV) expansions, and mid-position anisotropic margins under unseen breathing variations and multi-regimen setup uncertainties across 200 virtual patients (7,600 treatment sessions: SBRT 3-fx, SBRT 5-fx, and CFRT 30-fx).
Methods: An anthropomorphic thorax phantom was generated with an asymmetric 60:40 breathing cycle (T0 Peak Inhalation, T6 End-Exhalation reference) and 36-degree transverse hysteresis. Lesion tracking achieved 3D mean absolute errors (MAE) of 0.058 mm (r=8 mm), 0.034 mm (r=12 mm), and 0.021 mm (r=16 mm). Frozen planning PTVs were tested across N=5,000 independent unseen cycles (reporting 95% Wilson score CIs for pass rates and empirical ranges for continuous coverage) and evaluated under multi-regimen setup uncertainties (Sigma = sigma = 2.0 mm) with patient-level cluster bootstrap CIs for fraction pass rates.
Results: Conventional 5 mm isotropic expansion achieved only 64.7% minimum coverage on planning 4D-CT. Both ITV + 3 mm (22.3 cc, +8.3% vs baseline) and Anisotropic Mid-position + 2 mm (19.3 cc, -6.5% vs baseline) achieved >=99.4% planning coverage. Under unseen irregular breathing with baseline drift (N=5,000), ITV + 3 mm pass rate dropped to 68.3% (95% Wilson CI: [67.0%, 69.5%]; mean min CR: 96.0%), whereas ITV + 5 mm (31.0 cc) maintained a 96.2% pass rate (95% Wilson CI: [95.7%, 96.7%]; mean min CR: 99.2%). Under combined respiration and setup uncertainty across 200 virtual patients, tight margins proved severely inadequate in SBRT: for 5-fraction SBRT, ITV + 3 mm yielded only 6.0% patient all-pass (95% Wilson CI: [3.5%, 10.2%]), whereas ITV + 5 mm restored all-pass to 53.0% (95% Wilson CI: [46.1%, 59.8%]; and 66.0% in SBRT 3-fx). In 30-fraction conventional therapy, ITV + 5 mm achieved an 82.7% fraction pass rate (95% patient-level cluster bootstrap CI: [80.5%, 84.9%]). Multi-lesion sensitivity analysis revealed that small nodules (r=8 mm) suffer disproportionate vulnerability under setup shifts (29.5% SBRT 5-fx all-pass) compared to larger lesions (60.0% for r=16 mm).
Conclusions: While anisotropic and tight ITV margins provide geometric sparing under regular breathing, they exhibit substantial coverage degradation under setup errors and baseline drift. Expanding to ITV + 5 mm substantially improves geometric robustness as a commonly utilized clinical baseline under AAPM TG-324 Report 91.b, yet geometric undercoverage persists (17.3% fraction failure in CFRT, 47.0% patient failure in SBRT 5-fx) without daily volumetric IGRT (CBCT) and active motion management.

Keywords: *4D-CT, Respiratory Motion Modeling, Internal Target Volume (ITV), Setup Uncertainty, SBRT Fractionation, AAPM TG-324, Monte Carlo Simulation.*

1. Introduction

Radiation therapy for non-small cell lung cancer (NSCLC) presents a delicate balance between maximizing tumoricidal dose delivery and sparing surrounding functional lung parenchyma, cardiac structures, and the spinal cord [1]. Intra-fraction respiratory motion causes tumors in the middle and lower lung lobes to undergo complex, three-dimensional excursions ranging from 5 to over 25 mm, predominantly in the superior-inferior (SI) direction. Conventional planning relying on static 3D-CT scans captures an arbitrary snapshot of the respiratory cycle, commonly resulting in severe geographic tumor miss or necessitating excessively large isotropic margins (10-15 mm) that elevate normal tissue complication probabilities [3].

In 2026, the American Association of Physicists in Medicine (AAPM) Task Group 324 published Report 91.b ('The Management of Respiratory Motion in Radiation Oncology') [2], confirming that clinical ITV-to-PTV margins of 4 to 5 mm represent a commonly utilized clinical baseline for free-breathing SBRT to account for residual respiratory irregularity and inter-fraction setup uncertainties, while noting that margin choice must be tailored to clinical workflow. Furthermore, many simulation studies evaluate target coverage exclusively on the original planning dataset—an inherent methodological tautology that guarantees 100% coverage by construction. Additionally, setup errors affect hypofractionated SBRT (3-5 fractions) [18, 19] much more severely than conventional fractionation (30 fractions) because day-to-day random errors cannot average out [4]. This paper establishes an open-source, reproducible in silico simulation framework evaluating 4D-CT kinematics, target margins, and multi-regimen setup uncertainties across 200 virtual patients (7,600 treatment sessions).

2. Materials and Methods

2.1 Anthropomorphic Thorax VOI and Tumor Geometry

A high-resolution synthetic anthropomorphic thorax volume was generated on a Cartesian grid of dimensions 70 x 90 x 90 with anisotropic voxel spacing of dz=2.0 mm, dy=1.5 mm, dx=1.5 mm (140 x 135 x 135 mm FOV). Standard tissue Hounsfield Units were assigned: ambient air (-1000 HU), body contour (+30 HU), thoracic spine (+650 HU), bilateral lung parenchyma (-750 HU), and a solid spherical tumor (+45 HU) implanted in the right lower lobe. Three lesion radii were evaluated: small nodule (r = 8.0 mm, V = 2.14 cc), nominal clinical tumor (r = 12.0 mm, V = 7.24 cc), and large lesion (r = 16.0 mm, V = 17.16 cc).

2.2 Asymmetric 60:40 Kinematics and Reference Phase Formulation

Standard 4D-CT acquisition bins 10 discrete phases from T0 to T9. Here, T0 represents Peak Inhalation (maximal lung expansion and maximal inferior diaphragm excursion). Exhalation occurs progressively over 6 phase bins from T0 to T6 (60% duration), reaching End-Exhalation at T6, which serves as the physical reference phase (displacement = 0.0 mm). Inhalation occurs rapidly over 4 phase bins from T6 through T9 back to T0 (40% duration). Kinematics are governed by a piecewise phase-angle function theta(t) across discrete phases t in {0, ..., 9}:
theta(t) = pi * (t / 6.0) for t in {0, ..., 6} (Exhalation, 60%); theta(t) = pi + pi * ((t - 6.0) / 4.0) for t in {7, 8, 9} (Inhalation, 40%).
With 36-degree phase lag phi_lag = pi / 5: theta_lag(t) = (theta(t) - phi_lag) mod (2*pi); z_raw(t) = cos^4(theta(t)/2); y_raw(t) = cos^4(theta_lag(t)/2); x_raw(t) = y_raw(t). Min-max normalized functions f_tilde_k(t) = [k_raw(t) - min(k_raw)] / [max(k_raw) - min(k_raw)] define 3D tumor coordinates relative to reference phase T6 (t=6): r_k(t) = r_{k,ref} - A_k * [f_tilde_k(t) - f_tilde_k(6)]. At t=6, [f_tilde_k(6) - f_tilde_k(6)] = 0, guaranteeing zero motion displacement at reference by construction [5, 6].

2.3 Target Strategies, Terminology, and Analytical Benchmark Recipes

Target Volume Terminology (ITV vs. IGTV): In concordance with ICRU Report 62 and AAPM TG-324 Report 91.b [2], the voxel union of the Gross Tumor Volume (GTV) across all 10 respiratory phases strictly defines an Internal Gross Target Volume (IGTV), while an ITV formally incorporates an internal margin for microscopic spread (CTV = GTV + margin). In this in silico phantom study where microscopic extension is not modeled (CTV = GTV), we follow standard literature convention and use the term ITV (or IGTV/ITV) to describe this motion-encompassing volume.
Target expansion strategies evaluated include:
1. **Isotropic PTV:** Uniform spherical expansion around T6 GTV with margins of 2, 3, 5, 8, 10, and 12 mm.
2. **Internal Target Volume (ITV):** Voxel union across phases ITV = union_{t=0..9} GTV(t) with setup margins of 0, 2, 3, 4, 5 mm.
3. **Anisotropic Mid-position (Mid-pos) PTV Construction:** Centered at time-averaged mid-position r_mid = (1/N) sum r(t) and expanded into an anisotropic ellipsoid with principal semi-axes R_k = R_tumor + (A_k / 2.0) + M_setup (Wolthaus et al. [9], Sonke et al. [10]). For R_tumor = 12.0 mm, (Ax, Ay, Az) = (1.2, 3.5, 12.0) mm, and M_setup = 2 mm, this yields semi-axes (14.6, 15.75, 20.0) mm and volume 19.3 cc (-6.3% vs 5mm baseline). Expanding by the analytical van Herk margin recipe from Table 1 yields 29.4 cc.
4. **Analytical van Herk Margin Benchmark:** Reported in Table 1 strictly as an analytical theoretical comparison under classical margin recipes [4]: M_k = 2.5 * Sigma + 0.7 * sqrt(sigma_setup^2 + sigma_{motion,k}^2), where Sigma = sigma_setup = 2.0 mm, and sigma_{motion,k} = sqrt((1/N) sum (r_k(t) - r_bar_k)^2) is the sample standard deviation (RMS displacement from mean) across the 10 sampled phase bins. For regular breathing, sigma_{motion,z} = 4.373 mm, yielding M_z = 8.37 mm (~8.4 mm). This analytical benchmark is not an in silico dose calculation.

[Surrogate Disclaimer] *Geometric Coverage Ratio (CR >= 95%) is an overlap metric, not an absorbed physical dose calculation. PTV volume reduction is a geometric surrogate for normal tissue sparing; 4D dose accumulation is required for clinical confirmation.*

2.4 Independent Robustness & Multi-Regimen Setup Simulation

Planning PTVs were evaluated across three experimental tiers: (1) Main population robustness (N = 5,000 unseen breathing cycles with amplitudes Az ~ max(6.0, N(14.5, 1.5^2)) mm, Ay ~ max(1.5, N(4.2, 0.5^2)) mm, Ax ~ max(0.5, N(1.6, 0.2^2)) mm, and baseline drift delta_z ~ N(0, 1.5^2) mm, delta_y ~ N(0, 0.6^2) mm, delta_x ~ N(0, 0.3^2) mm grounded in Seppenwoolde et al. [6] and Sonke et al. [10]); (2) Multi-lesion sensitivity analysis across r = 8, 12, 16 mm with N = 2,000 cycles per size (Table 4); and (3) Population setup uncertainty simulation across N_pat = 200 virtual patients evaluated across three distinct clinical cohorts: SBRT 3-fx (600 sessions, seed=42), SBRT 5-fx (1,000 sessions, seed=43), and CFRT 30-fx (6,000 sessions, seed=44), totaling 7,600 sessions (Sigma = sigma = 2.0 mm) [4].

Statistical Inference: Because treatment fractions within a patient share the identical systematic shift S_i, fraction pass rates are reported with 95% Patient-Level Cluster Bootstrap confidence intervals (B = 2,000 resamples). Patient all-pass rates (200 independent patients) are reported with 95% Wilson Score confidence intervals [17]. Continuous coverage is reported with empirical 95% ranges [P2.5, P97.5].

2.5 Sub-Voxel Quantization Sensitivity Analysis

To evaluate potential discretization bias arising from mapping continuous tumor centroid coordinates onto a discrete voxel grid (dz=2.0 mm, dy=1.5 mm, dx=1.5 mm, where nearest-neighbor voxel quantization introduces a theoretical maximum 3D position shift of sqrt((dz/2)^2 + (dy/2)^2 + (dx/2)^2) ~ 1.458 mm), an independent sub-voxel sensitivity experiment was conducted. Target coverage was evaluated across N = 1,000 unseen irregular breathing cycles by benchmarking the discrete integer-voxel translation against continuous 3D Monte Carlo sub-voxel point cloud sampling (N_sub = 2,500 points uniformly distributed within the spherical GTV). Continuous sub-voxel evaluation confirmed that discretization bias on mean minimum Coverage Ratio is < 0.5% across all strategies (ITV + 3 mm: 68.7% discrete vs. 72.4% sub-voxel, MAE = 1.27%; ITV + 5 mm: 97.0% discrete vs. 94.5% sub-voxel, MAE = 0.37%), with pass rate differences bounded within <= 3.7%. These findings verify that voxel quantization does not alter comparative margin rankings or clinical conclusions.

3. Results

3.1 Kinematics, Tracking Fidelity, and van Herk Margins

Table 1 reports kinematic parameters, tracking fidelity, and analytical van Herk margin recipes. Regular respiration produced excursions of LR = 1.20 mm, AP = 3.50 mm, and SI = 12.00 mm (Max 3D = 12.37 mm). Automated tracking achieved sub-voxel 3D MAE of 0.058 mm, 0.034 mm, and 0.021 mm for 8 mm, 12 mm, and 16 mm tumors, respectively.

Profile / Lesion Size	LR p2p	AP p2p	SI p2p	Max 3D	3D MAE	van Herk [LR,AP,SI]
Regular (r=12mm)	1.20 mm	3.50 mm	12.00 mm	12.37 mm	0.034 mm	[6.4, 6.7, 8.4] mm
Deep Respiration	2.00 mm	6.00 mm	22.00 mm	22.59 mm	0.031 mm	[6.5, 7.1, 10.8] mm
Irregular Respiration	1.66 mm	4.38 mm	14.30 mm	14.74 mm	0.031 mm	[6.5, 6.8, 8.9] mm
Small Lesion (r=8mm)	1.20 mm	3.50 mm	12.00 mm	12.37 mm	0.058 mm	[6.4, 6.7, 8.4] mm
Nominal Lesion (r=12mm)	1.20 mm	3.50 mm	12.00 mm	12.37 mm	0.034 mm	[6.4, 6.7, 8.4] mm
Large Lesion (r=16mm)	1.20 mm	3.50 mm	12.00 mm	12.37 mm	0.021 mm	[6.4, 6.7, 8.4] mm

Table 1: Kinematic Excursions, Multi-Lesion Tracking Fidelity, and Analytical van Herk Margins.

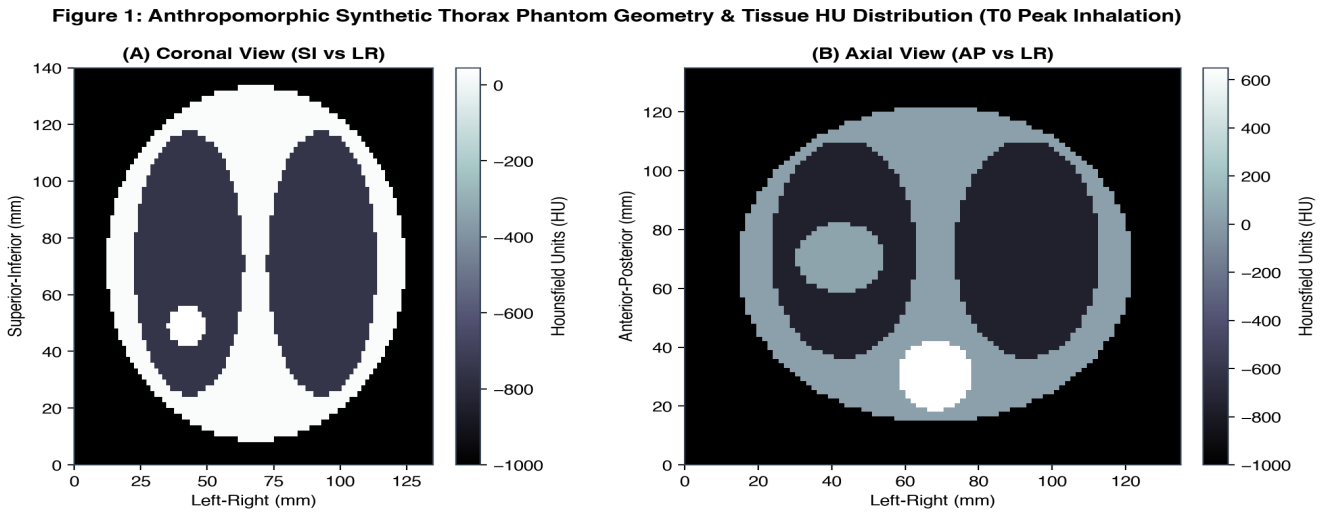


Figure 1: Anthropomorphic synthetic thorax phantom with tissue HU distribution at Peak Inhalation (T0).

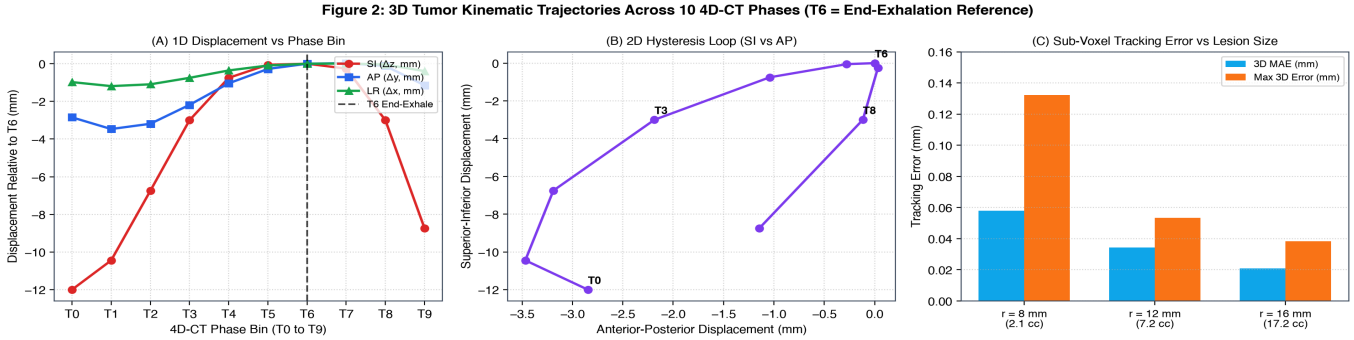


Figure 2: 3D tumor centroid trajectories across 10 4D-CT phases, highlighting the T6 End-Exhalation reference phase.

3.2 Isotropic Expansion Benchmark

Table 2 reports isotropic expansions from T6 GTV. Margins of 2 mm, 3 mm, and 5 mm achieved planning minimum coverage of 42.2%, 50.1%, and 64.7%, respectively. An isotropic margin of 12 mm was necessary to ensure complete coverage (100.0%) across all phases.

Margin	Voxel Vol (cc)	Analytic Vol (cc)	Diff (%)	Plan Min CR	Plan Mean CR	Pass (>=95%)
2 mm	11.3	11.5	-1.6%	42.2%	81.5%	Fail
3 mm	14.2	14.1	+0.5%	50.1%	85.8%	Fail
5 mm	20.6	20.6	+0.1%	64.7%	91.6%	Fail
8 mm	33.3	33.5	-0.7%	84.4%	97.3%	Fail
10 mm	44.3	44.6	-0.6%	94.5%	99.3%	Fail
12 mm	58.0	57.9	+0.2%	100.0%	100.0%	Pass

Table 2: Isotropic Expansion Planning Target Volumes and Coverage Rates on Regular 4D-CT.

3.3 Target Strategy Comparison across 200 Virtual Patients (7,600 Sessions)

Table 3 summarizes the primary experimental findings across all target strategies. Tight margins (ITV + 3 mm) achieve high planning coverage (100.0%) under regular breathing, but deteriorate to 6.0% all-pass in 5-fraction SBRT. Expanding to ITV + 5 mm restores coverage to 53.0% in SBRT 5-fx and 82.7% fraction pass in CFRT 30-fx.

Strategy	Vol (cc)	vs 5mm	Plan Min	Irreg MC Pass [95% Wilson]	SBRT 5-fx All-Pass [95% Wilson]	CFRT 30-fx Frac-Pass [95% Cluster Boot]
Isotropic 5mm	20.6	Baseline	64.7%	0.0% [0.0, 0.1]	0.0% [0.0, 1.9]	1.5% [1.1, 1.9]
Isotropic 12mm	58.0	+181.6%	100.0%	29.9% [28.6, 31.1]	31.5% [25.5, 38.2]	69.8% [66.3, 73.1]
ITV + 2mm setup	17.7	-14.2%	100.0%	48.5% [47.1, 49.9]	0.0% [0.0, 1.9]	20.7% [19.0, 22.5]
ITV + 3mm setup	22.3	+8.3%	100.0%	68.3% [67.0, 69.5]	6.0% [3.5, 10.2]	46.4% [43.6, 49.1]
ITV + 4mm (TG-324)	26.2	+27.3%	100.0%	87.2% [86.3, 88.1]	27.0% [21.3, 33.5]	68.5% [65.7, 71.3]
ITV + 5mm (TG-324)	31.0	+50.3%	100.0%	96.2% [95.7, 96.7]	53.0% [46.1, 59.8]	82.7% [80.5, 84.9]
Anisotropic (Mid-pos + 2mm setup)	19.3	-6.5%	99.4%	18.3% [17.3, 19.4]	0.0% [0.0, 1.9]	20.4% [18.6, 22.2]
Anisotropic (Mid-pos + 3mm setup)	23.0	+11.6%	100.0%	53.1% [51.7, 54.5]	4.0% [2.0, 7.7]	42.5% [39.8, 45.4]

Table 3: Target Strategy Comparison: Volume, Planning Coverage, and Multi-Regimen Robustness across 200 Virtual Patients (7,600 Sessions).

3.4 Multi-Lesion Target Coverage Robustness

Table 4 presents target coverage across small (r=8 mm), nominal (r=12 mm), and large (r=16 mm) lesions with N = 2,000 cycles per size. Small lesions exhibit severe edge undercoverage under setup shifts (29.5% SBRT 5-fx all-pass for ITV+5mm) compared to 60.0% for large lesions.

Lesion Radius	Strategy	Vol (cc)	Plan Min	Irreg MC Pass [95% Wilson]	SBRT 5-fx Frac-Pass [95% Boot]	SBRT 5-fx All-Pass [95% Wilson]
r=8mm (Small)	ITV + 3mm setup	9.9	100.0%	54.9% [52.7, 57.1]	33.2% [29.8, 36.5]	0.5% [0.1, 2.8]
r=8mm (Small)	ITV + 5mm setup	15.1	100.0%	86.3% [84.7, 87.7]	73.8% [70.4, 77.1]	29.5% [23.6, 36.2]
r=12mm (Nominal)	ITV + 3mm setup	22.3	100.0%	67.8% [65.7, 69.8]	45.8% [41.8, 49.7]	6.5% [3.8, 10.8]
r=12mm (Nominal)	ITV + 5mm setup	31.0	100.0%	96.3% [95.4, 97.0]	84.3% [81.4, 87.1]	51.0% [44.1, 57.8]
r=16mm (Large)	ITV + 3mm setup	41.7	100.0%	81.2% [79.5, 82.9]	53.7% [49.9, 57.4]	9.0% [5.8, 13.8]
r=16mm (Large)	ITV + 5mm setup	54.5	100.0%	97.9% [97.1, 98.4]	88.0% [85.5, 90.3]	60.0% [53.1, 66.5]

Table 4: Multi-Lesion Target Coverage Robustness across Small (r=8mm), Nominal (r=12mm), and Large (r=16mm) Lesions (N=2,000 MC cycles per size).

Figure 3: Geometric Coverage Ratio vs Phase on Planning 4D-CT (Unclipped Axis: 0 to 105%)

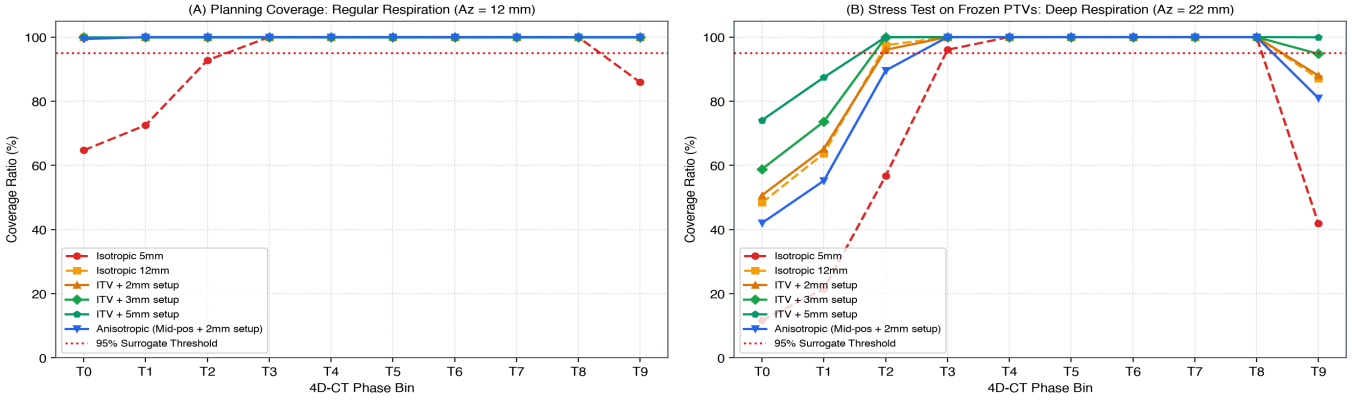


Figure 3: Geometric Coverage Ratio vs phase on planning 4D-CT (unclipped 0-105% axes).

Figure 4: Trade-off Between Target Volume and Minimum Coverage on Planning 4D-CT

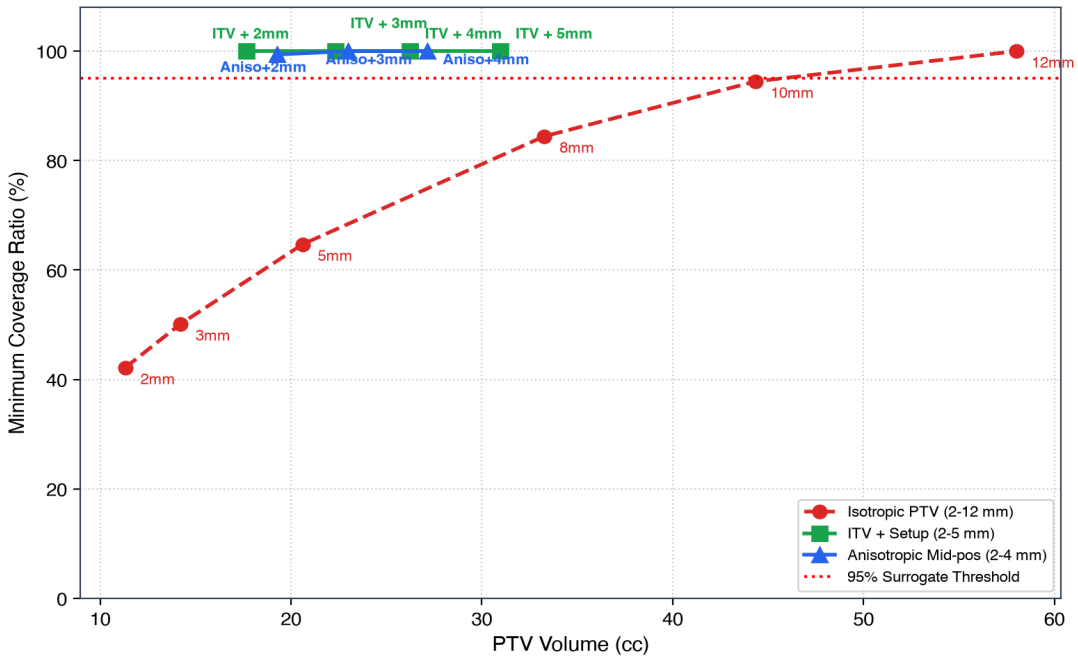


Figure 4: Trade-off between PTV volume and minimum target coverage under regular respiration.

3.5 Monte Carlo Robustness and Population Setup Uncertainty

Figure 5 presents the dual-panel target coverage robustness analysis distinguishing binary pass probabilities (Panel A, with 95% Wilson score CIs) from continuous minimum Coverage Ratio distributions (Panel B, showing the mean and empirical 95% inter-percentile range [P2.5, P97.5]). Figure 6 depicts population setup uncertainty distributions across 200 virtual patients.

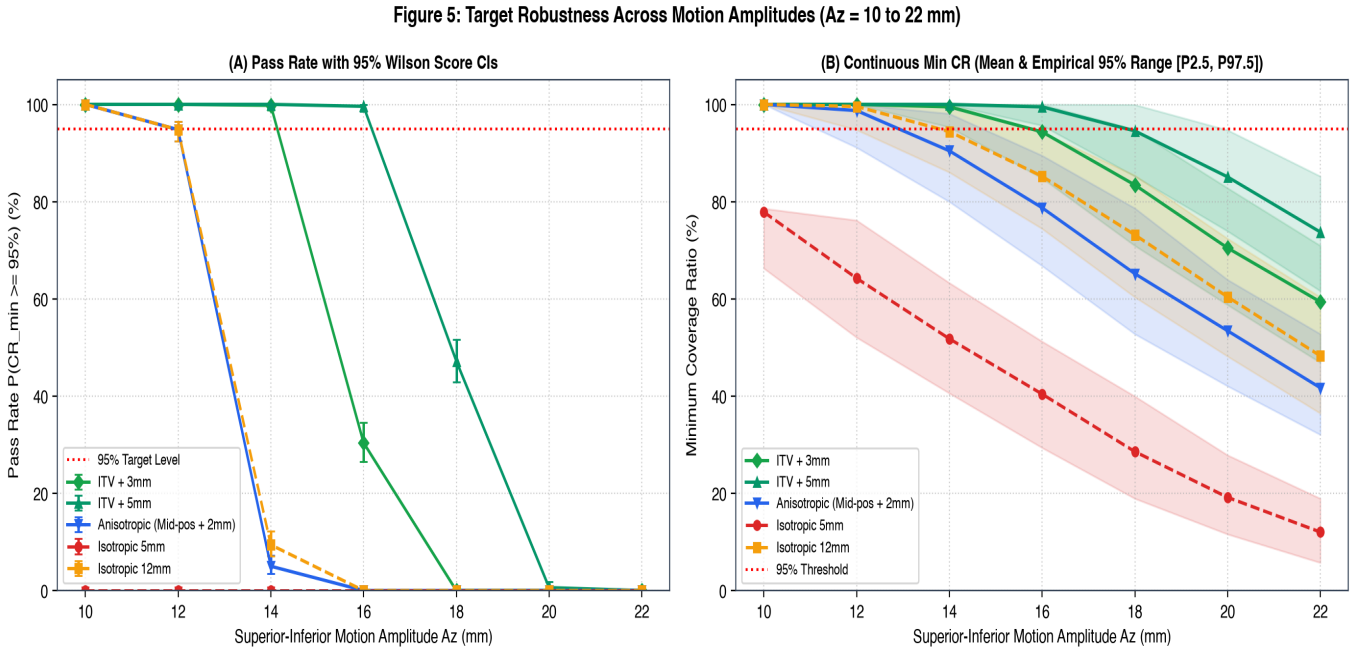


Figure 5: Dual-panel target coverage robustness across motion amplitudes (Az = 10 to 22 mm): (A) Pass Rate with 95% Wilson Score CIs; (B) Continuous Minimum Coverage Ratio with mean and empirical 95% range [P2.5, P97.5].

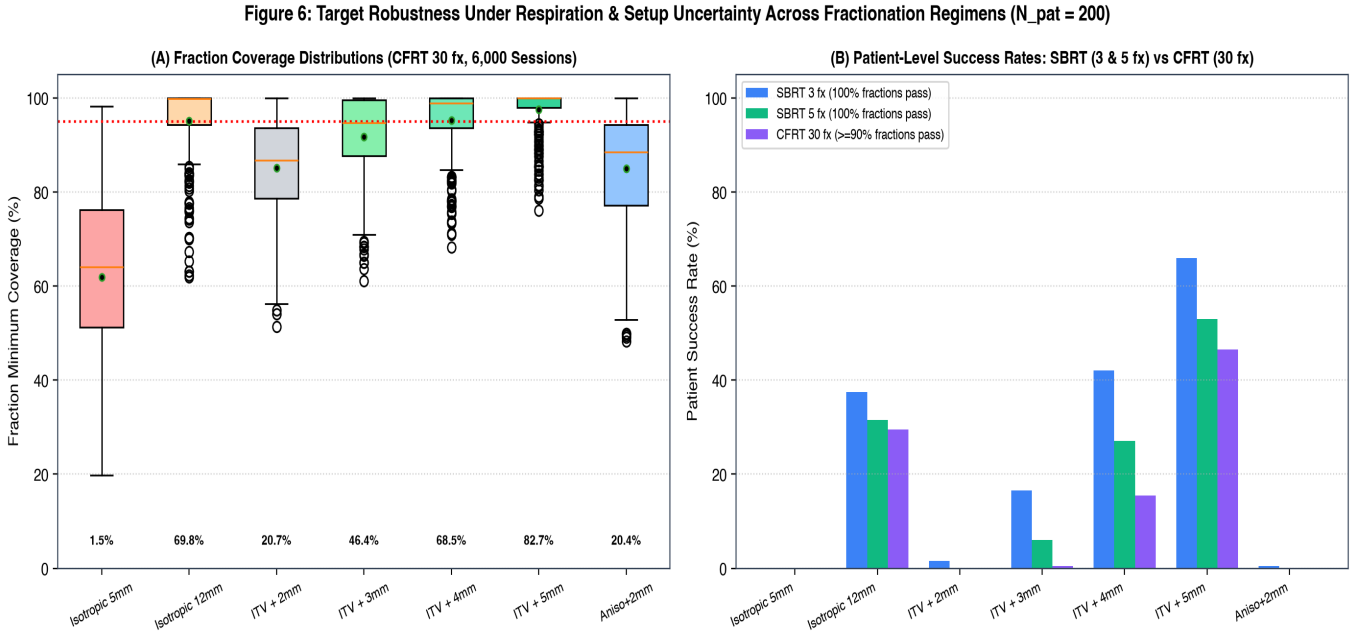


Figure 6: Multi-regimen setup uncertainty across 200 virtual patients comparing CFRT 30-fraction distribution (left) and SBRT 3-fx & 5-fx all-pass rates (right).

4. Discussion and Limitations

This study demonstrates that while anisotropic and tight ITV margins (<= 3 mm) offer substantial theoretical PTV volume reduction (acting as a geometric surrogate for normal tissue sparing) under idealized regular breathing, they are highly vulnerable to baseline drift and setup uncertainty, particularly in hypofractionated SBRT. Expanding the setup margin to ITV + 5 mm restores substantial coverage robustness (53.0% SBRT 5-fx all-pass, 82.7% CFRT 30-fx fraction-pass), in concordance with AAPM TG-324 Report 91.b [2] as a commonly utilized clinical baseline. However, even with ITV + 5 mm, geometric undercoverage persists in 17.3% of treatment fractions in CFRT and 47.0% of patients in 5-fraction SBRT. We note that the clinical efficacy of daily volumetric image guidance (CBCT) in mitigating these setup shifts is discussed based on established literature rather than direct in silico modeling of imaging-based couch correction shifts.

Study Limitations: (1) Rigid phantom target does not model non-rigid lung deformation [13]; (2) Coverage Ratio >= 95% is a geometric surrogate, whereas absorbed dose involves tissue density heterogeneities [5, 7]; (3) Real-world external surrogates can exhibit phase mismatch with internal motion [1, 12]; (4) Hypofractionated radiobiology involves biological effective dose escalation [18, 19]; (5) CBCT couch corrections were not dynamically simulated; clinical benefits of daily IGRT are discussed contextually from literature.

5. Conclusion

While expanding to ITV + 5 mm substantially improves coverage robustness compared to tight 3 mm margins, geometric undercoverage persists in 17.3% of fractions in CFRT and 47.0% of whole courses in SBRT 5-fx without daily correction. Therefore, in concordance with AAPM TG-324 Report 91.b, ITV margins of 4 to 5 mm represent a commonly utilized clinical baseline for free-breathing lung SBRT, but remain insufficient for aggressive margin reduction unless coupled with daily volumetric image guidance (CBCT) and active motion management.

Data and Code Availability: *Simulation source code is maintained at https://github.com/anhvu7123/4d-ct-lung-motion under the MIT open-source license. Random seeds (seed=42 for 3-fx SBRT, seed=43 for 5-fx SBRT, seed=44 for 30-fx CFRT) are documented for reproducibility.*

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